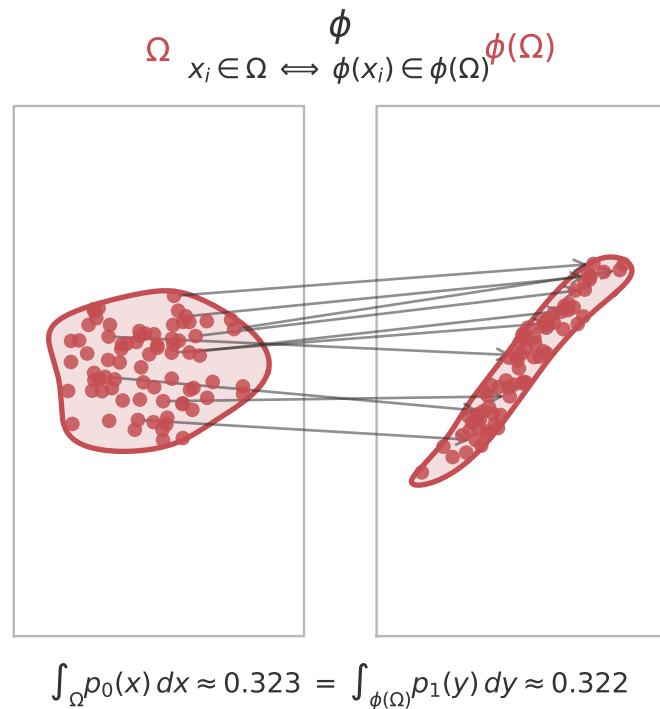
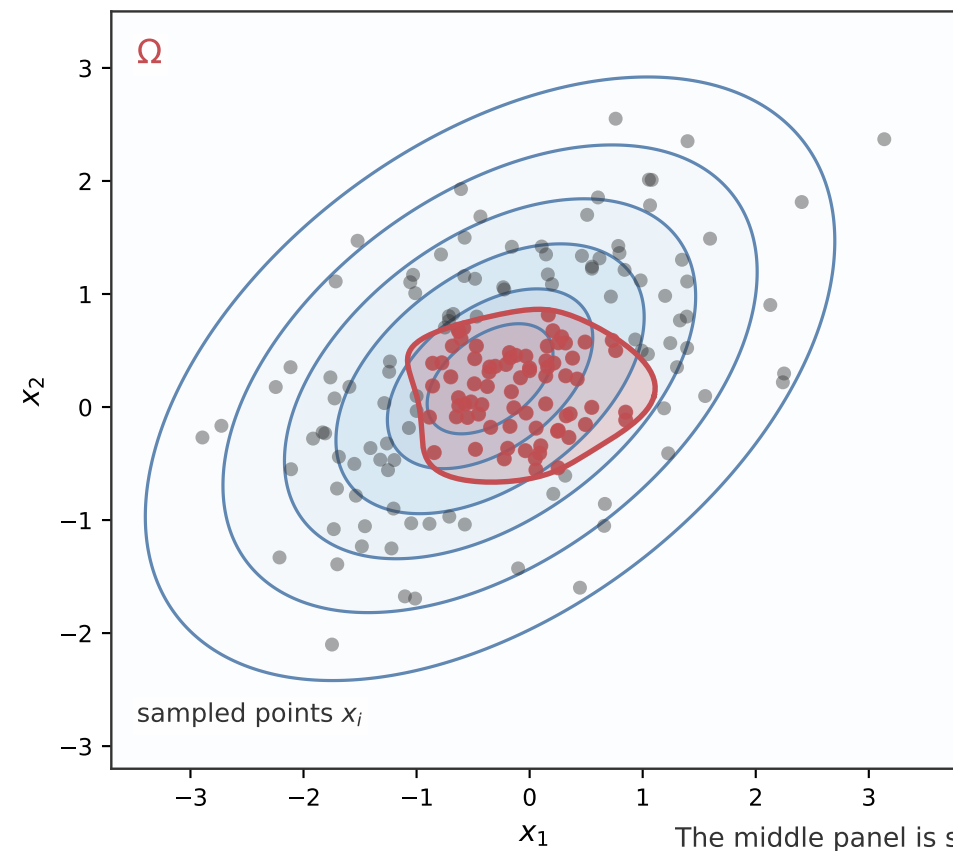


Moving density and conservation of mass under a diffeomorphism

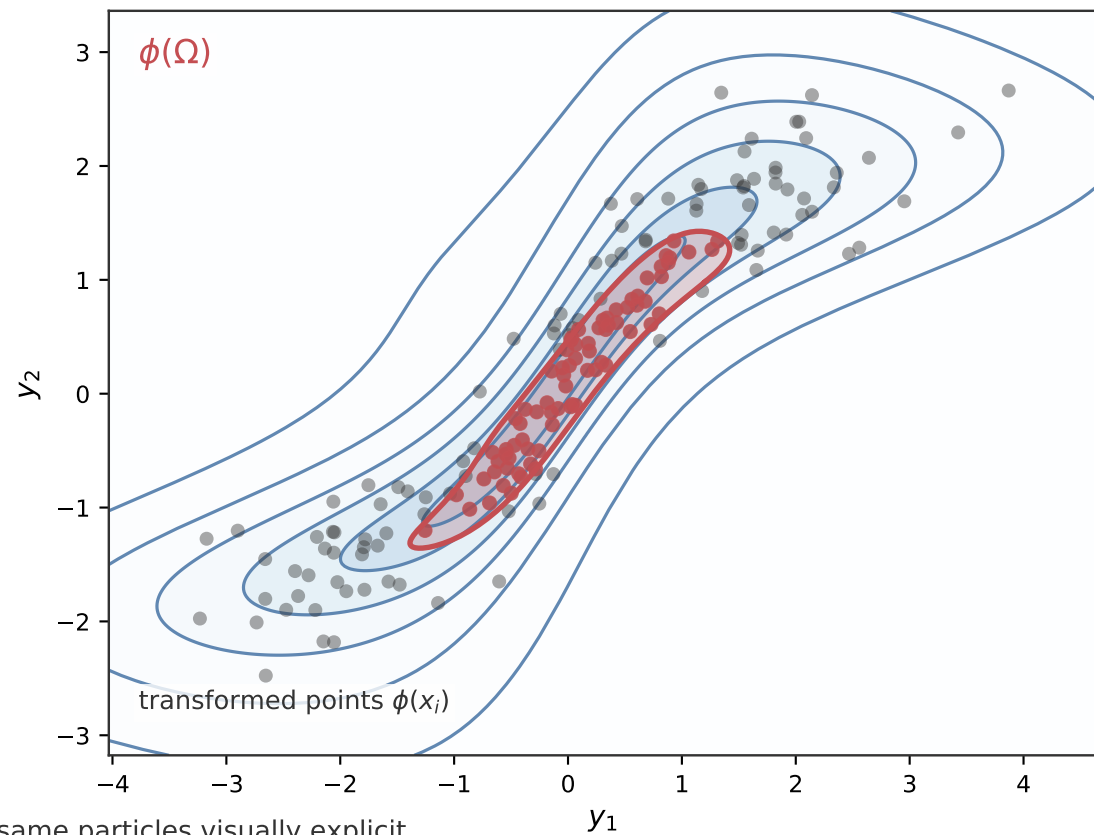
$$p_1(y) dy = \phi_{\#}(p_0(x) dx) \quad \Rightarrow \quad \int_{\phi(\Omega)} p_1(y) dy = \int_{\Omega} p_0(x) dx$$

Same material region, same probability mass

Initial density p_0 and subset Ω



Pushforward density $p_1 = \phi_{\#}p_0$



The middle panel is schematic: it places Ω and $\phi(\Omega)$ side by side to make the transport of the same particles visually explicit.